200 Generated Smart Buildings: Feature Extraction Report

NeurIPS 2026 Smart Buildings starter kit | generated with the kit's own floor and scenario generators

1. Summary

The notebook **generate_200_buildings.ipynb** was run on Kaggle and produced **200 distinct buildings** (one row each, 54 columns) in **buildings_200_features.csv**, plus **feature_catalog.csv** (133 rows) and **rejected_seeds.csv** (82 seeds). All 14 validation checks in the notebook passed: 200 rows, unique seeds, no duplicate configurations or floor-plan arrays, no missing values, and the structural checks on floors, zones and AHUs. Observation *values* are not in the CSV because they exist only during a simulation rollout; they are listed in the catalog as simulation-dependent.

2. Method

Two-stage generation. Stage 1 (`run_floor_generator`) turns a seed into a building YAML. Stage 2 (the scenario generator, same functions as `run_scenario_generator.py`) turns that YAML plus one weather preset into `floor_plan.npy`, `zone_map.npy` and `config.gin`. One fixed preset (`atlanta_1`) is used for all buildings; it is needed only to run stage 2.

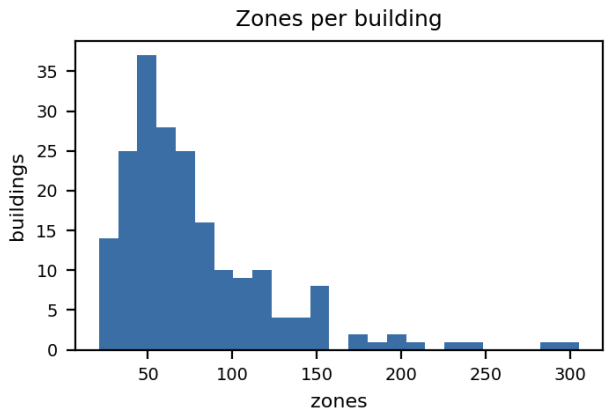
Seeds. Seeds run upward from 100000 until 200 are accepted (last accepted: 100281). The generator itself rejects a seed when any zone is under 20 m^2; **82** of 282 seeds tried were rejected and are logged, not hidden.

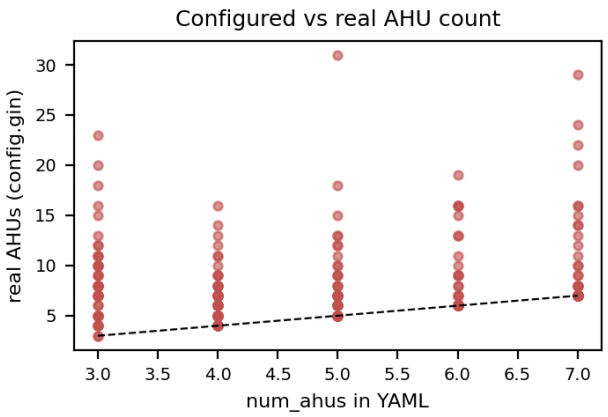
Zone count. A zone is a connected region of value 0 in `zone_map.npy`, cross-checked against the room lists in `config.gin` for every building. On the shipped examples both give 30, 39 and 85 zones. The earlier documented 28 for building_train_8 disagrees with both and was not used.

Observation schema. Field names come from constructing the real environment for the three shipped buildings. The vector length is **6 x zones + 15 x AHUs + 58** (6 per VAV zone, 15 per AHU, 7 hot-water, 48 time, 3 building-wide). It reproduces 283, 352 and 703 on the shipped buildings, matched the real environment on three generated buildings (smallest, median and largest zone count), and holds in all 200 CSV rows. The numbers 42 and 47 are not the vector length.

3. Results

Quantity	Min	Median	Max
Zones per building	21	64	305
Real AHUs (from config.gin)	3	7	31
AHUs configured in YAML	3	5	7
Floors	2	3	4
Observation vector length (derived)	229	547	2353





The real AHU count exceeds the configured one in **146 of 200** buildings, because the scenario generator adds handlers when a floor plan has more rooms than the per-AHU cap (10) allows. The CSV stores both counts. Every building is a composite footprint with the architectural style and a 30-day episode, so those columns do not vary in this sample.

4. Output files

File	Content
buildings_200_features.csv	200 rows x 54 columns: meta (4), cfg (22, static building config), derived (16, structure from arrays and gin), obs (12, schema counts)
feature_catalog.csv	133 rows: the CSV columns plus 79 simulation-dependent observation feature types (6 zone, 15 AHU, 7 hot-water, 48 time, 3 building)
rejected_seeds.csv	82 seeds rejected by the generator's zone-area check, with the reason

5. Caveats

- The observation length for the other 197 generated buildings is *derived* by the rule above, not checked against a live environment each time.
- The generated buildings are larger than the six shipped examples (up to 305 zones and 31 AHUs), following the generator's own width and height ranges.
- Seeds and the weather preset were my choices, not competition rules. The weather preset should not change the floor plan or zone map, based on reading the code; this was not tested.
- Zone-to-AHU assignment is in `config.gin`; the CSV keeps only the minimum and maximum rooms per AHU.